Design and Actuation of Conformal Magneto-Mechanical Metamaterials

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Background and Motivation

Metamaterials are engineered materials with unique mechanical and wave manipulation properties enabling rapid shape changes. Magneto-mechanical metamaterials, controlled through magnetic fields, achieve property tuning via shape reconfiguration. Hard-magnetic soft materials, created by embedding hard magnetic particles into a soft polymeric matrix, undergo complex, untethered, reversible, and rapid deformations in response to magnetic fields.[1]

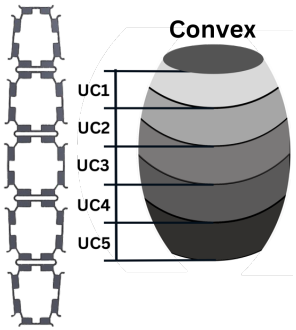
This research focuses on hard-magnetic soft materials, investigating 3D substrate structures such as concave, convex, and cylindrical shapes. We perform experimental tests and simulations to explore how these materials interact with electromagnetic fields. Additionally, we study the deformation under different amperages and concentrations of magnetic powder.

Previous research shows that 2D magneto-metamaterials allow for tunable performance through electrical/mechanical reconfiguration strategies.[2] demonstrating the broader versatility of metamaterials.[2] Through manufacturing and designing 3D metamaterial substrates similar tests can be performed highlighting the importance of continued investigation into magnetic deformation.

3D Design

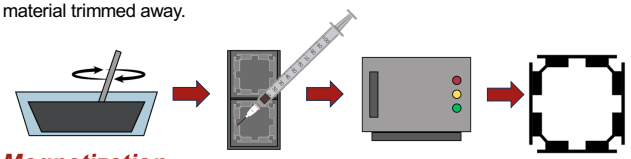
Unique unit cells are designed for each metamaterial section based on the specific 3D shape. While the height of the cells remains constant, the width and side angles vary to ensure proper conformation to the 3D surface. This customization is essential for the array to fit correctly and function effectively.

For example, the cylindrical structure uses cells with uniform dimensions and connection points to maintain its shape. In contrast, the concave and convex structures require cells with adjusted widths and side angles to match their respective curvatures.



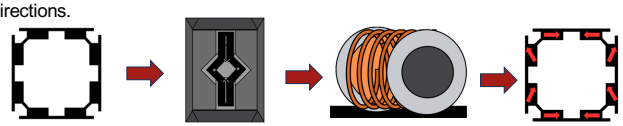
Fabrication

The existing metamaterial design and fabrication process developed by the lab is used to test all magneto-metamaterials. The unit cells are fabricated to ensure uniformity and functionality. We prepared a Dragon Skin 20 silicone elastomer blend and NdFeB magnetic particles. This mixture is degassed and then poured into 3D-printed molds made from Polyvinyl alcohol (PVA) filament. The molds are heated at 80ºC for 30 minutes to cure the material. Post-curing, each unit cell is carefully extracted or dissolved from the molds, with any excess material trimmed away.

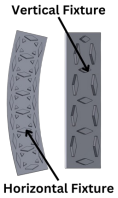


Magnetization

Following the fabrication of the unit cells, they are positioned in a 3D-printed fixture that is arranged in a diamond-like pattern. These cells are placed inside a 1.5 T magnetic impulse field to establish the predetermined magnetization directions.



Assembly

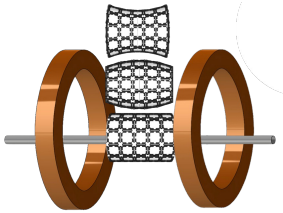


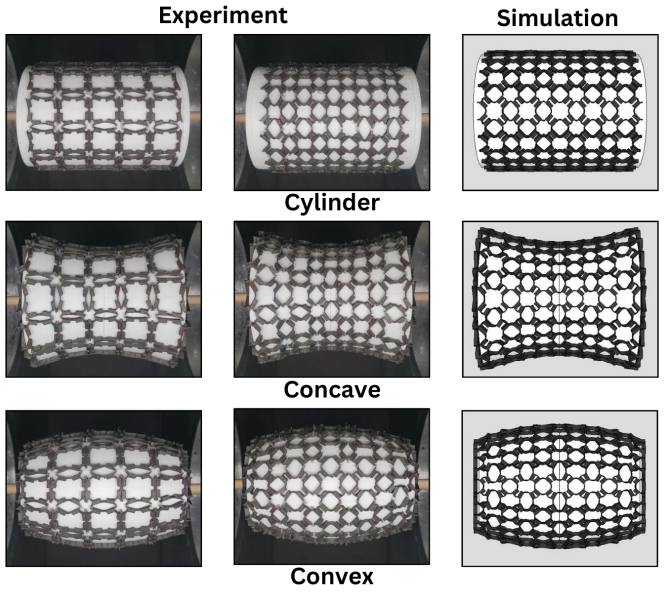
Once magnetized, the array is assembled by connecting the hinges of the unit cells with silicone epoxy. The cells are first connected horizontally, then vertically. Using the UC cell labeling system, UC1 cells are assembled horizontally, and then UC1-UC5 arrays are connected vertically.

We use uniform cells with precise connection points throughout the cylindrical configuration. Connections are made at angles to best fit the curved 3D shapes for the concave and convex configurations.

Cylinder, Concave, Convex Actuation

The first actuation involves metamaterials in concave, convex, and cylindrical shapes, each with a 12×5 grid of cells containing 25% magnetic powder. A magnetic field of 50 mT is applied to each test. Two magnetic coils are placed on either side of the substrate to actuate these structures.





The same manufacturing methods and actuation procedures are used for the convex, concave, and cylindrical structures. The unit cells are shaped and angled differently to match each substrate's curvature, allowing the metamaterials to better conform to the 3D shapes. Results demonstrate the feasibility of creating functional metamaterials that adapt to curved edges. This finding highlights the potential of metamaterials to achieve precise deformation patterns and conform to various geometries.

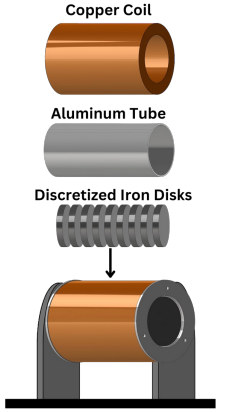
Simulation

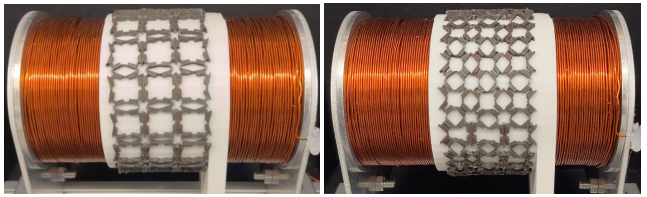
The cell arrays and substrates underwent simulation and experimental testing processes to ensure accuracy. Simulations using ABAQUS 2021 involve detailed meshing of each cell partition. The cell arrays are then experimentally tested with the coils. The experimental results consistently match the simulation predictions, confirming the precision of our models.

Discrete Iron Core Electromagnetic Coil

The hollow circular structure, made of 16×3 cells, is placed around a magnetic coil with a cast iron coil. The unit cells contain 35% magnetic particles and are actuated using a current range of 10-22.5A. Despite the iron core coil typically being stronger than an ironless coil, it is the smallest in this setup. The magnetic particle percentage is increased to balance the magnetic field interactions and achieve even actuation, similar to the actuation in the convex, concave, and cylindrical substrates.

The iron coil is made up of three important components. First, the copper coil is evenly wound against the aluminum tube and then discretized iron disks are evenly distributed along the inside core of the coil. These components combine to create an even magnetic field.





Deployed Folded

At 22.5A, peak actuation is achieved, showing the effectiveness of the magnetic field in inducing deformation. The iron core field strength impacts the responsiveness of the magneto-metamaterial. The increased percentage of magnetic particles allows for more pronounced deformation, highlighting the importance of particle concentration in achieving the desired actuation. These results show that fine-tuning the magnetic particle concentration and applied current, combined with the iron core, can optimize magneto-metamaterial performance, providing insights into the design of future systems.

Future Studies & Conclusion

The study demonstrates the potential of magnetically actuated metamaterials across various 3D substrates, influenced by the unit cell design. Results show that optimizing magnetic particle concentration and applied current is crucial for achieving desired deformation patterns. Future research should focus on testing the efficiency of magneto-metamaterials tunable performance through electrical/mechanical reconfiguration.[2] Additionally, exploring new design and manufacturing approaches for unit cell configurations and overall structure can enhance metamaterial performance. The field is still developing, with opportunities for further investigation into manufacturing techniques, design methodologies, and practical applications.

Acknowledgments

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References

[1] Lu, L., Sim, J., & Zhao, R. R. (2024). Mechanics of hard-magnetic soft materials: A Review. *Mechanics of Materials*, *189*, 104874. https://doi.org/10.1016/j.mechmat.2023.104874
[2] Wu, S., Eichenberger, J., Dai, J., Chang, Y., Ghalichechian, N., & Zhao, R. R. (2022). *Supporting Information for "magnetically Actuated Reconfigurable Metamaterials as Conformal Electromagnetic Filters.* https://doi.org/10.22541/au.165478179.98256479/v1